Chapter 1

THE PROBLEM AND ITS BACKGROUND

1.1 Introduction

Soil liquefaction is a geotechnical phenomenon that occurs when saturated, loose soils lose their strength and behave like fluids during an earthquake. Ground shaking increases pore water pressure while decreasing effective stress, hence reducing soil particle interaction. This soil fabric degradation causes ground settlement, lateral spreading, and structural failure (Seed & Idriss, 1971).

Liquefaction not only produces an immediate loss of shear strength, but it also results in significant post-liquefaction settlement when excess pore pressures decrease and the soil reconsolidates. According to studies, once the shaking stops, the quick rise and subsequent dissipation of pore pressure cause the soil structure to collapse, resulting in vertical deformations that can last a long time after the earthquake. According to analytical and experimental studies, reconsolidation settlement can be big enough to destroy foundations, pavements, and underground lifelines, especially when structures are underlain by thick liquefiable layers (Chiaradonna, et al., 2019). Recent centrifuge and shake-table studies show that many existing numerical models are unable to adequately represent the magnitude and rate of these settlements, underscoring the complicated and damaging character of post-liquefaction ground deformation (Kontoe et al., 2022). Because such settlements can cause differential displacement, tilting, reduced bearing capacity, and even structural failure, the aftermath of liquefaction poses significant risks to infrastructure. This reduction in bearing capacity, which refers to the soil's ability to support foundation loads, is a direct result of the loss of soil volume and structural integrity during and after pore pressure dissipation, emphasizing the importance of accounting for liquefied soil's long-term reconsolidation behavior.

Several major earthquakes throughout the world have highlighted the damaging effects of soil liquefaction, making it a critical issue in geotechnical and seismic engineering research. The 1964 Niigata earthquake (Mw 7.5) in Japan is one of the best-documented events, with significant liquefaction of loose, water-saturated sands causing ground subsidence, lateral spreading, and the collapse of multiple structures and bridges (Ishihara & Koga, 1981). Similarly, in New Zealand, the 2010–2011 Canterbury earthquake sequence triggered extensive liquefaction in river-adjacent and deltaic deposits, severely damaging residential infrastructures and lifeline systems (Cubrinovski et al., 2011; Cubrinovski et al., 2012). These global case studies emphasize the importance of reliable liquefaction prediction methods, which have evolved from empirical and probabilistic approaches to advanced data-driven solutions.

Traditional liquefaction evaluation was mostly based on Seed and Idriss' (1971) simplified method, which relied on empirical correlations between field data and cyclic resistance ratios. Traditional techniques, on the other hand, usually fail to capture the complex, site-specific relationships between soil qualities, seismic stress, and groundwater conditions, making it difficult to predict liquefaction potential efficiently. Liquefaction evaluation has evolved alongside advances in computing technology, including the introduction of machine learning and artificial intelligence techniques. Bayesian Belief Networks and Decision Trees have been shown to outperform traditional empirical methods in accounting for multidimensional variable interaction (Muftuoglu &

Dehghanian, 2025). BiLSTM, a deep learning framework, has achieved above 95% predictive accuracy, indicating its potential to improve hazard prediction (Şehmusoğlu et al., 2025). Furthermore, recent breakthroughs in liquefaction research demonstrate the real-world applicability of Machine Learning (ML) and Artificial Neural Networks (ANNs), especially when verified against earthquake case histories. Models are trained on large global datasets created by events such as the Niigata, Kobe, Chi-Chi, and Canterbury earthquakes, and then tested on previously unexplored locations to ensure excellent generalization performance (Zhu et al., 2017). Their utility is further proved when used to recent seismic occurrences, such as the 2023 Türkiye earthquakes, where ML models correctly predicted liquefied zones using post-event geotechnical data (Sanger et al., 2024). Furthermore, in cities such as Yokohama, Japan, ANN-based hazard maps have been incorporated into planning projects to guide land use and risk mitigation (Rai et al., 2024). Similarly, Geographical Information System (GIS) have proven play a crucial role in regional hazard assessment by allowing the integration of multiple environmental and spatial layers. GIS-based morphotectonic analysis has been effectively used to delineate liquefaction susceptibility zones over extensive geological regions, demonstrating the capability of spatial tools to identify potential hazards at a regional scale (Rawat et al., 2022). In urban environments, GIS has also been applied to assess social vulnerability by combining liquefaction susceptibility mapping with socio-economic data to support disaster resilience and risk reduction planning (Manoharan & Ganapathy, 2023). Overall, these applications highlight the strength of GIS in providing clear spatial visualization of risks across both regional and urban context.

Despite the sophistication of these computational methods, previous studies that relied solely on ML or GIS provided valuable insights, but their capabilities were frequently limited ML-only models can detect complex patterns in liquefaction data but lack spatial context, whereas GIS based mapping provides clear spatial visualization but is heavily reliant on static, empirical inputs. Importantly, both models fail to assess post-liquefaction consequences because they do not forecast settlement or examine the resulting loss of soil bearing capacity, both of which are necessary for recognizing actual infrastructure risk. Combining ML with GIS provides researchers with a more comprehensive framework for quantitative prediction and spatial visualization. This study extends this framework by implementing a web-based architectureutilizing PostGIS for efficient spatial data management and Leaflet for real-time interactive mapping, allowing for a more accessible and dynamic assessment of geotechnical risks.

Nevertheless, post-liquefaction ground settling is still relatively unknown, even though it poses a considerable risk to infrastructure performance and safety. Recent studies have begun to address this gap through data-driven methods. For example, Park, Ogunjinmi, Woo, and Lee (2020) developed an Artificial Neural Network (ANN)-based model for predicting settlement at the Pohang earthquake site, demonstrating that machine learning can effectively quantify the often-complex volumetric reconsolidation of liquefied soils. This method provides a useful tool for engineering applications by allowing preliminary settlement estimation with limited field data such as Standard Penetration Test (SPT) findings and cycle stress ratios. However, the study emphasizes that settlement prediction is highly site-specific, dependent on local soil stratigraphy, groundwater conditions, and seismic loading characteristics, emphasizing the critical need for additional research in post-liquefaction settlement, particularly in regions like the Philippines where earthquake-prone, liquefiable soils are common (Park et al., 2020).

As part of the Pacific Ring of Fire, the Philippines experiences frequent seismic activity and possesses geological conditions that make it susceptible to liquefaction. This vulnerability was highlighted by the magnitude 7.4 Davao Oriental earthquake on October 10, 2025, which caused lateral spreading and ground deformation in coastal zones with saturated sandy soils, emphasizing the recurring nature of liquefaction-related ground failures in seismically active regions (PHIVOLCS, 2025). Historical records further support this risk: a national database documenting over 808 liquefaction incidents from 1619 to 2020 indicates that areas which experienced liquefaction during past earthquakes remain highly vulnerable to future events, particularly those situated on reclaimed lands, coastal zones, and river deltas (Buhay et al., 2024). Probabilistic mapping studies, utilizing over 1,000 borehole logs and local seismic data, have revealed that several densely populated areas—such as the Pasig River floodplains and reclaimed Manila Bay—are exposed to moderate to high liquefaction risk, especially under magnitude 7.2 earthquake scenarios along the West Valley Fault (Dungca, 2016).

This study aims to develop an online tool that predicts soil liquefaction potential, settlement and bearing capacity in Tarlac Province. By integrating an ANN with PostGIS and Leaflet, the system will generate spatial susceptibility maps and provide site-specific foundation recommendations. The tool's methods and engineering standards, ensuring its reliability for resilient structural planning and disaster risk reduction.

1.2 Objectives of the Study

The main objective of this research is to develop an ANN web-based application with GIS mapping to predict soil liquefaction potential, settlement, bearing capacity and provide foundation recommendations that support safer structural planning in liquefaction-prone areas of Tarlac. Specifically, this study aims to address the following:

1. Collect geotechnical and seismic hazard parameters to be organized in MS Excel.
2. Development and training of the ANN model that predicts soil liquefaction potential and its impact on settlement and bearing capacity.
3. Validate the predictive performance of the ANN model output by comparing it to the following:
    a. Liquefaction Screening Potential based on DPWH BSDS (2013)
    b. Tokimatsu & Seed (1987) Method
    c. Bray & Macedo (2017) Method
    d. Terzaghi's (1943) Bearing Capacity Theory
    e. Olsen & Stark (2002) Method
4. Integrate the predicted ANN model output into a web-based application with GIS mapping.
5. Recommend foundations based on liquefaction susceptibility, quantifying the resulting impact on soil bearing capacity and total settlement.

1.3 Significance of the Study

This study aims to develop a web-based prediction system that assesses soil liquefaction potential, its impact on settlement and bearing capacity, and provide foundation recommendations in Tarlac Province. The system is intended to support engineers, planners, and local government units in making safe and data-driven decisions for infrastructure development. This study offers a highly accurate, data-driven, and location-specific prediction tool by collecting and organizing geotechnical and seismic parameters, developing and validating an Artificial Neural Network (ANN), and integrating the outputs into a web-based application with GIS mapping to visualize liquefaction susceptibility, settlement, bearing capacity, and recommended foundation types.

1.4 Scope and Delimitations

The study focuses on developing an ANN model to predict soil behavior within the Province of Tarlac. The research aims to provide predictive tools that support preliminary structural assessment. The study involves a layer-by-layer check of soil data to estimate settlement and bearing capacity, and recommend suitable foundation, based on the susceptibility of the site.

The study is limited up to a depth of 15 meters, where the ANN model performs a dual-pathway analysis to determine the most suitable foundation. For liquefiable soils, where $FS \leq 1.0$, the ANN predicts both the free-field vertical settlement, and differential settlement, and the post-liquefaction bearing capacity. For non-liquefiable soils, where $FS > 1.0$, the system estimates the ultimate bearing capacity and total settlement.

To ensure the accuracy of the ANN's predictions, the model's outputs are validated through the established engineering methods, including DPWH BSDS (2013) standards for the liquefaction screening, Tokimatsu & Seed (1987) and Bray & Macedo (2017) methods for settlement analysis, and Terzaghi's (1943) Theory and Olsen & Stark (2002) method for the bearing capacity evaluation.

The use of GIS in this study is limited to mapping the spatial distribution of liquefaction susceptibility across Tarlac Province. The system uses Leaflet as the front-end engine display that allows users to easily see the results generated by the ANN model. Only boreholes with reliable input data – such as SPT as unit weight (γ), corrected SPT blow count ($N_{1(60)}$), cyclic stress ratio (CSR), angle of internal friction (φ), and cohesion (c) – are included in the modeling process. Boreholes lacking these parameters are excluded to maintain consistency and reliability of the ANN models. Also, the study is limited to undeveloped or vacant lands and does account for sites with existing structures. Additionally, it does not cover the lateral spreading, or detailed soil-structure interaction modeling. All outputs serve as screening-level information and should be supplemented with comprehensive investigation for actual design or construction purposes.

1.5 Conceptual Framework

This research lies in an integrative approach that combines data-driven machine learning models, spatial data processing, and engineering decision-support tools. The framework is designed to enable end-to-end processing: from raw data acquisition and preprocessing, through risk prediction, to actionable recommendation and spatial framework visualization.

Figure 1

Conceptual Framework

INPUT

Borehole logs
a. Coordinates
b. SPT results
c. Depth
d. Lithology

Soil Index Properties
a. Grain Size Analysis
b. Atterberg Limits
c. Soil Density

Seismic Hazard Parameters
a. Peak Ground Acceleration
b. Earthquake Magnitude
c. Fault Distance

Spatial and Environmental Data
a. Geographic Coordinates
b. Digital Elevation Models

PROCESS

Data Processing and Management
a. Data digitization and cleaning
b. Normalization of SPT data

Machine Learning Development
a. Feature extraction and engineering
b. Artificial Neural Network Training
c. Model Validation (RMSE and Confusion Matrices)

Geospatial Processing
a. Borehole Logs Digitization
b. PostGIS Processing

System Development
a. Web-based Interface Integration

OUTPUT

ANN-Web Application
a. Liquefaction Potential
b. Settlement of Soil
c. Foundation Recommendation
d. Interactive Maps

Decision Support Tool
a. Scenario Simulation and Reporting

Datasets and open-access models for further research/practice

1.6 Review of Related Literature and Studies

Spatial Mapping of Soil Liquefaction

Soil liquefaction mapping is a systematic process of identifying and delineating areas susceptible to soil liquefaction. The mapping is accomplished using Geographic Information System, a computational tool used to convert raw data into usable map that support evidence-based interpretation. These maps classify areas based on their susceptibility to liquefaction. This concept is significant as it enables the transformation of site-specific data into hazard map. The hazard map serves as mandatory preliminary screening tool used to delineate high-hazard zones.

The evolution of methods and tools for mapping of soil liquefaction has been demonstrated for the past studies. Wang et al. (2023) used high-resolution aerial imagery into a GIS framework. This process allowed for the creation of Digital Surface Models (DMSs) and Digital Orthophoto Maps (DOMs), which facilitated the recognition of 39,286 individual liquefaction features. The study demonstrated the GIS's ability to produce rapid, large-scale hazards maps.

Buhay et al. (2024) developed a comprehensive database of historical liquefaction occurrences in the Philippines, documenting 808 accounts from years 1619 to 2020. The study identifies Tarlac as a region of highly susceptibility, particularly triggered by the extensive ground failures recorded during the 1990 Luzon Earthquake. The consolidation of centurial records establishes a "hazard memory" that provides a critical foundation for spatial mapping.

Haifani et. al (2023) evaluated different GIS interpolations methods, including Kriging and Inverse Distance Weighting (IDW), and found that methodological choices significantly affect the apparent spatial distribution of liquefaction-prone areas. Similarly, Karakas et. al (2024) integrated Artificial Intelligence (AI) predictive models with GIS to visualize the complex datasets, emphasizing the role of machine learning in enhancing earthquake resilience for smart cities.

GIS-based approaches offer the advantage of handling large, heterogeneous datasets and producing spatially continuous hazard maps. While other studies improved the predictive accuracy, their outputs remain static assessments that identify risk zones. Methodologies are delimited by their focus on surficial manifestations and qualitative records, often lacking the geotechnical parameters required for precise engineering. Current tools are often descriptive and inaccessible for structural planning. This study addresses these deficiencies by integrating the geotechnical parameters with an ANN model to ensure that the spatial mapping serves as a dynamic, computational tool.

Integration of Machine Learning Models in Predicting Soil Liquefaction

Soil liquefaction is a geotechnical phenomenon in which saturated, loose soils temporarily lose their strength and stiffness during seismic activity. Traditional empirical methods, such as SPT-based charts and factor of safety calculations, the complex, nonlinear interactions among soil properties, groundwater conditions, and seismic parameters are often not fully captured. Therefore, Artificial Neural Networks (ANN) have been adopted as a valuable tool in predicting soil liquefaction, as intricate patterns from multi-dimensional datasets can be learned and adaptive, data driven predictions can be generated.

Historically, the foundations for the modern ANN applications have been established by classical research. Tokimatsu and Seed (1987) introduced an SPT-based methodology for estimating settlements due to liquefaction, in which the importance of CSR and stress reduction factors was emphasized. Building upon this, post liquefaction ground deformations were quantified by Ishihara and Yoshimine (1992), and correlations between soil density, stress history, and volumetric strain were established. Furthermore, probabilistic approaches and liquefaction severity indices were proposed by Lee (2006-2007), which enhanced reliability-based assessment and continue to be referenced for validating ANN models. These classical studies provide parameters and benchworks that continue to guide contemporary computational modeling.

In recent years, advancements have been made in the application of ANN and other ML approaches in liquefaction prediction. Li et. al (2025) to develop a probabilistic ANN model based on 311 SPT case histories to estimate liquefaction likelihood. Their findings demonstrated that the ANN could better capture nonlinear relationships among geotechnical and seismic parameters compared to traditional empirical charts, particularly in site with heterogeneous soil profiles. Similarly, Pham et al. (2021) implemented feedforward neural networks using SPT data to predict liquefaction triggering. This study emphasized the significance of carefully selected input feature, including soil resistance and site-specific seismic measurements, and revealed that neural networks could offer timely and reliable assessment for engineering purposes. Galupino and Dungca (2023) applied ML algorithms, including ANNs, to generate a liquefaction susceptibility map for Metro Manila, Philippines. By integrating archived borehole and SPT data with site-specific parameters, their work demonstrated how local geotechnical records could be leveraged to produce hazard maps for urban planning and disaster preparedness. Collectively, these studies underscore the growing efficacy of ML in addressing the complexities of liquefaction prediction.

Methodologically, a structured workflow has been commonly adopted in these studies. Data collection and preprocessing are performed first, ensuring that inputs such as corrected SPT N-values, fine content, groundwater depth, peak group acceleration (PGA), and cyclic stress ratios (CSR) are reliable. Subsequently, appropriate neural network architectures, such as multilayer perceptron or convolutional networks, are selected according to dataset complexity and output requirements. Thereafter, hyperparameter optimization, cross-validation, and model evaluation using metrics such as accuracy, root-mean-square error (RSME), or area under the curve (AUC) are carried out to enhance reliability. In several cases, model performance is benchmarked against conventional empirical methods or alternative ML algorithms to contextualize predictive capability and assess practical relevance (Li et al., 2025; Pham et al., 2021; Galupino & Dungca, 2023).

These applications indicate that ML-based models can effectively account for nonlinear interactions among soil and seismic variables, enabling rapid and site-specific predictions. In addition, integrating ANN outputs with probabilistic and spatial mapping frameworks facilitates hazard visualization and planning, which is particularly valuable in densely populated regions such as Metro Manila. Nevertheless, outcomes are highly dependent on the data quality and representativeness, limited or biased datasets may compromise prediction consistency. Interpretability of neural network predictions remains a concern, occasionally limiting the adoption of ML tools in practical engineering contexts. Moreover, transferring trained models to regions with distinct soil and seismic conditions often requires retaining or adaptation, highlighting the importance of careful model calibration and local validation.

Despite these advancements, several research gaps remain. High-quality, region-specific datasets combining SPT, CPT, Shear-wave velocity, and site-specific seismic data are still scarce, restricting model generalizability and reproducibility. Additionally, approaches to uncertainty quantification and explainable AI are underutilized, which can hinder confidence in predictions. This study aims to address these gaps by integrating ANN with GIS. By combining ANN's capability to model complex, nonlinear soil-seismic interactions with GIS-based spatial analysis. This study will provide site-specific, interactive predictions of liquefaction susceptibility, settlement, and residual bearing capacity.

Free-Field Ground Settlement

Free-field ground settlement or earthquake-induced settlement is a downward movement of the ground surface in an area without structures, typically caused by earthquake-induced soil liquefaction. Where saturated sandy soils are subjected to a sequence of cyclic shear stresses caused by earthquake loading, leading to the build-up of excess pore water pressure. The subsequent dissipation of this excess pore pressure results in a reconsolidation of volumetric strain. The magnitude of this reconsolidation volumetric strain is influenced by factors such as the sand's grain size, its relative density, the maximum shear strain induced in the deposit, and the amount of excess pore pressure generated by the earthquake. The significance of this concept for this research lies in the fact that the magnitude of post-earthquake settlement, determined by factors such as sand relative density and excess pore pressure generation, directly affects bearing capacity in regions such as Tarlac.

In a related advancement, the practical application of ANN models has been demonstrated in South Korea, where researchers used machine learning to predict liquefaction-induced settlement in the Heunghae Basin near Pohang City following the Mw 5.4 earthquake of November 2017. The study developed an Artificial Neural Network (ANN) model using Standard Penetration Test (SPT)–based parameters to estimate earthquake-induced settlement. Two ANN models were evaluated: Model 1, which used unit weight (γ), corrected SPT blow count (N1(60)), and cyclic stress ratio (CSR), and Model 2, which used depth (d), N1(60), and CSR. Results showed that Model 1 achieved higher predictive accuracy, with an R^2 of 0.8601 and a Mean Absolute Error (MAE) of 0.1941, compared to Model 2 with an R^2 of 0.7352, indicating a difference of about 0.12 in R^2. Sensitivity analysis further revealed that unit weight (γ) had the strongest influence on predicted settlement, while N1(60) contributed more significantly than CSR. Using the optimal model, the authors proposed a simplified predictive equation and a liquefaction-induced settlement chart based on relationships among N1(60), CSR, and settlement. The applicability of this chart was confirmed through comparisons of predicted settlement values with measured ground deformation obtained from Interferometric Synthetic Aperture Radar (InSAR) imaging. Overall, the simplified ANN-based settlement chart provided engineers with a practical and efficient tool for estimating liquefaction-induced settlement using minimal input parameters, offering a more streamlined alternative to complex analytical or numerical procedures (Park et al., 2020)

Over the past two decades, various procedures have been developed to study the earthquake-induced settlement problem, ranging from complex non-linear dynamic computer models (like DESRA or TARA3FL) to simplified procedures. The most widely recognized simplified approach was developed by Tokimatsu and Seed (1987), who identified the primary factors controlling settlement as the cyclic stress ratio (CSR) combined with pore pressure generation, the corrected Standard Penetration Test N-value (SPT-N), and the earthquake magnitude. Their methodology utilizes an empirical chart to estimate the volumetric strain after liquefaction for a magnitude 7.5 earthquake directly from the cyclic stress ratio and normalized SPT-N value. Alternatively, Ishihara and Yoshimine (1992) developed a graphic representation as a methodology to evaluate post-liquefaction volumetric strain as a function of the factor of safety against liquefaction, maximum shear strain, and relative density. A more simplified and practical methodology was later proposed by C. Y. Lee (2006-2007), which is based on the Tokimatsu and Seed (1987) procedure but replaces the cumbersome charts and diagrams with a set of direct equations. This equation-based approach allows for convenient numerical implementation and the expeditious analysis of complicated, multi-layered soil profiles.

A primary advantage of traditional methods is their reliance on the Standard Penetration Test (SPT). The criteria for evaluating liquefaction behavior using SPT are regarded as robust, supported by extensive databases compiled from sites that have experienced liquefaction. The approach by Tokimatsu and Seed (1987) is well known and widely used in practice. However, a key weakness of the original Tokimatsu and Seed (1987) approach is that it "often requires the use of charts, tables or diagrams," making it time-consuming, especially when analysing sites with multiple soil layers. The simplified equation-based approach (Lee, 2006-2007) successfully addresses this weakness by facilitating numerical implementation and expediting the analysis of complicated soil profiles. Validation of this simplified method showed that its settlement predictions were in reasonably good agreement with field measurements from two major seismic events: the Tokachi-Oki earthquake (1968) and the Loma Prieta earthquake (1989), sometimes yielding results closer to the measured settlement than the Tokimatsu-Seed approach. Despite these promising results, a significant gap remains: the simplified volumetric strain lines used in this new approach are based on limited data points. Consequently, the observation regarding the liquefaction boundary line "may or may not be valid in cases other than those in the data set analyzed". Therefore, further validation of the proposed approach using additional field liquefaction data from earthquakes with various magnitudes is warranted to strengthen its reliability.

Liquefaction-Induced Building Settlements

One of the major challenges in earthquake engineering practice is liquefaction-induced settlement beneath shallow foundations. This response is commonly interpreted through three (3) primary methods: volumetric-induced, ejecta-induced and shear-induced. Simplified estimation procedures have become prevalent for volumetric-induced settlement (Δv), which primarily caused by post-liquefaction one-dimensional (1D) reconsolidation under free field conditions. Furthermore, these established empirical methods, such as those by Ishira and Yoshimine (1992) as well as Zhang et al. (2002) capture settlement related to soil sedimentation and reconsolidation. In contrast, the shear-induced settlement that originates from Soil-Structure Interaction (SSI)-induced such as cyclic ratcheting and bearing capacity failure, is usually responsible for the severe structural damage, however, this method historically lacked simplified estimation approaches. Consequently, effective evaluation of building performance requires consideration of all settlement mechanisms, highlighting shear-induced deformation, which frequently dominates the total settlement behavior.

To address the complexity of shear-induced settlement, Bray & Macedo (2017) developed a simplified procedure derived from more than 1,300 Nonlinear Dynamic SSI effective stress analyses. Their approach identifies and incorporates critical sites, structural, and seismic parameters through numerical simulations. Critical site parameters influencing settlement are characterized by the relative density (D_r) and thickness of the liquifiable layer (H_L), which are collectively represented through Liquefaction-Induced Building (LBI) index. The LBI index relates to cyclic shear strain potential (ε_{Shear}) of liquified soils, typically estimated using CPT-based methods like those from Zhang et al. (2002). Structural effects are accounted for through parameters including building foundation contact pressure (Q) and the foundation width (B), while seismic demand ground motion characteristics described using standardized cumulative absolute velocity (CAV_{dp}) and the 5%-damped one-second spectral acceleration (Sa).

The simplified procedure developed by Bray & Macedo (2017) was validated using field case histories, centrifuge test data, and analytical investigation, including observations from the 2010-2011 Canterbury earthquake sequence. For instance, the application of the simplified method to building C3 in Adapazari, Turkey yielded a median estimated total settlement that closely matched the observed minor settlement. In addition, advanced modeling technique have been investigated to supplement simplified analytical approaches. Chithuloori and Kim (2025) developed machine learning-based prediction models, specifically Random Forest (RF) and Multilayer Perceptron (MLP) ANN, using 1-G shaking table experimental data. Their research highlighted the importance of incorporating parameters such as pore water pressure and acceleration, improves prediction performance with RF model achieving a reported accuracy of 98.87% for liquefaction-induced settlement. While simplified procedure provides reasonable estimates that are consistent with advanced numerical analyses, the highly complex nature of the phenomenon requires that the results should be complemented with project-specific Nonlinear Dynamic SSI effective stress analyses.

Determination of Soil Bearing Capacity

Soil bearing capacity refers to the greatest load that the earth can safely hold without shear failure or excessive settlement, making it an important concern in foundation design. Terzaghi's classical theory was one of the first and most influential methods for estimating bearing capacity. It provides a simplified yet reliable approach for shallow foundations, particularly square and rectangular footings, by analyzing the shear failure surface at the foundation's base (Terzaghi, 1943). Despite its simplicity, Terzaghi's approach is still commonly employed in conventional geotechnical practice due to its conservative estimations, low data needs, and demonstrated reliability. Later developments included shape, depth, and load-inclination parameters (Meyerhof, 1963). Additional corrections for soil compressibility, sloping ground, and inclined footing bases have been introduced to provide more realistic analysis of complex foundation conditions and different geological settings (Hansen, 1970; Vesic, 1975).

Contemporary research demonstrates the breadth of computational and empirical approaches to measuring soil bearing capacity. For example, multiple design perspectives have been used to assess bearing capacity across Ile-Ife (Alawode et al., 2020), while comparative investigations using IS code methods, Terzaghi's formulation, and Geo5 software highlight how outputs can differ across analytical frameworks. However, many of these systems require substantial geotechnical inputs or consider a wide range of footing shapes, which are not always available. Given these limits, Terzaghi's technique remains ideal for contexts with limited data and square or rectangular footings, such as those used in this study. Its conservative forecasts and uncomplicated computation make it suitable for use with the geotechnical parameters available in Tarlac Province.

This practicality is also evident in regional studies that used comparable methods to characterize shallow soil holding capacity. Researchers in Metro Manila estimated allowed capacities across the region using corrected SPT N-values and traditional bearing capacity theories such as Terzaghi and Vesic's. The examination of 486 boreholes and GIS-based contour maps indicated predictable changes in soil strength, including low-capacity alluvial zones along coastal and low-lying areas (0-200 kPa), and larger capacities in central regions underlain by the thick Guadalupe Formation (≥300 kPa). These findings demonstrate that SPT-based approaches, particularly Terzaghi's simplified approach, are feasible, conservative, and reliable for locations with varied soil conditions and insufficient subsurface data, making them suitable for usage in contexts such as Tarlac Province (Dungca et al., 2017).

While traditional theories provide solid baseline estimations, they presume pre-liquefaction soil conditions and fail to account for the liquefaction-induced loss in soil strength. Soils can face severe shear strength deterioration after cyclic loading and excess pore pressure building, resulting in post-liquefaction settlement and reduced bearing capacity. To solve this issue, post-liquefaction assessment methods have been developed, the most notable being Olsen and Stark's methodology. Olsen and Stark developed the Liquefied Strength Ratio (LSR) by back-analyzing 33 reported liquefaction flow failures. The LSR relates the residual shear strength of liquefied soils to the effective overburden stress before to failure. Their method, which correlates residual shear strength with in-situ data such as SPT and CPT, provides a useful foundation for assessing post-liquefaction bearing capacity. (Olsen & Stark, 2002).

Terzaghi's approach is utilized in this study to determine the pre-liquefaction baseline bearing capacity, which is then adjusted using the Olsen & Stark framework based on the site's estimated liquefaction potential. This integrated technique considers both the initial soil strength and post-liquefaction deterioration, allowing for a realistic and safe assessment of foundation performance in Tarlac Province's liquefaction-prone locations. By combining conventional and post-liquefaction approaches, the study provides a systematic framework for predicting foundation behavior under seismic loads and providing practical design suggestions for disaster-resilient infrastructure. (Terzaghi, 1943; Olsen & Stark, 2002).

Integration of ANN in Predicting Soil Bearing Capacity

In recent years, Artificial Neural Networks (ANNs) have emerged as an effective tool in geotechnical engineering for modeling complicated, nonlinear interactions between soil parameters, seismic stress, and foundation performance. ANNs may learn from past records to anticipate outcomes such as soil bearing capacity, settlement, and liquefaction-induced ground deformation, even when standard analytical methods are constrained by assumptions or insufficient data. This data-driven technique enables engineers to combine several interdependent elements at the same time, increasing forecast accuracy and allowing for site-specific assessments. Studies have effectively used ANN models to predict liquefaction-induced settlement in Pohang, Korea, indicating how machine learning can supplement traditional bearing capacity evaluations and post-liquefaction evaluation (Park et al., 2020). The Pohang study shows how ANNs can make rapid, accurate, and site-specific predictions of liquefaction-induced settlement, which can then be combined with classical and post-liquefaction frameworks to improve foundation design.

Similarly, the practical application of ANN models has been established in the Philippines, where they have been applied to predict soil bearing capacity in regions with limited geotechnical data. A study in San Fernando and Santo Tomas, Pampanga used an ANN model to estimate allowable bearing capacities based on corrected SPT N-values, N60, friction angle, unit weight, and footing width, with initial calculations based on Terzaghi's bearing capacity formula. The model demonstrated high accuracy, with a correlation coefficient near 1 and very low mean squared error and identified unit weight and N60 as the most influential parameters. By integrating ANN predictions with GIS-based spatial analysis, engineers were able to rapidly assess site-specific soil bearing capacities, highlighting the practical advantage of machine learning in supplementing traditional foundation design methods where field data are sparse (Lingad et al., 2023).

Overall, the use of ANNs in soil carrying capacity and post-liquefaction research is a big step forward in geotechnical engineering. ANN models can capture complex, nonlinear relationships between soil parameters, seismic stresses, and groundwater conditions that are typically difficult to portray using traditional analytical approaches alone. This feature enables faster, more precise, and site-specific prediction of liquefaction-induced settlement and residual bearing capacity, which streamlines the evaluation and design process. ANN-based modeling, when combined with classical methods and post-liquefaction frameworks, provides a comprehensive, practical, and data-driven approach to evaluating foundation performance in liquefaction-prone areas, allowing for safer and more efficient structural planning (Terzaghi, 1943; Olsen & Stark, 2002; Park et al., 2020).

Foundation Selection on Liquefiable Ground

A foundation serves as an essential structural component that transfer loads from the structure above the foundation into the soil beneath. Its importance becomes crucial in liquefiable ground conditions, where saturated cohesionless soil lose shear strength and act like a viscous liquid during seismic events. Bray and Dashti (2010) indicate that liquefaction leads to ground displacement, including volumetric strain and soil softening which may lead to severe structural failure. The selection of appropriate foundation serves as a mitigation to ensure the serviceability and safety of the structure.

Current research emphasizes a hierarchical methodology to foundation selection based on the severity of ground deformation. According to Bachman (2015), the use of shallow foundation, including isolated footing, tend to be limited to locations where ground movement is minimal. Seed et al. (2001) indicate that settlements on liquifiable soil are typically manageable in the range of 3cm to 5cm if the post-liquefaction factor of safety is greater than 2.0. When these safety thresholds are compromised, mat or raft foundations are suggested as a more economical alternative. Ishii and Tokimatsu (1988) determined that when a width of foundation is at least two to three times the thickness of the liquefiable layer, the resulting building settlement will approximately equal the free field ground settlement. Ayadat (2017) demonstrated through a case study involving foundation design for an electrical station on a liquefiable site in Ontario. Designs must ensure that the total settlement does not exceed 25 mm and differential settlement remains below 20 mm to prevent structural misalignment. In addition, ASCE 7-16 and the DPWH BSDS (2013) set specific thresholds that for high-risk sites, lateral spreading limits between 4 to 18 inches beyond which designers use of deep foundation (piles).

The integration of data-driven methods into foundation recommendation remains underdeveloped. Existing foundation recommendations often focus on settlement, with less consideration given to the effect of bearing capacity degradation. The case study demonstrated that no single solution is universally applicable. Addressing the gaps, this study aims on developing region-specific frameworks that combine empirical knowledge, analytical modeling, and machine learning, incorporating bearing capacity degradation, settlement, and spatial variability into a unified system.

Synthesis and Justification

Soil liquefaction is recognized as a significant geotechnical hazard in earthquake-prone areas, as it can induce ground instability, leading to settlement and lateral spreading. The determination of soil liquefaction is commonly conducted through field investigation, such as the Standard Penetration Test (SPT), and the collection of geotechnical and seismic data. However, the availability of these data is typically limited at sites where specific projects are conducted, which prevents liquefication assessment from being applied to other locations.

Samui and Sitharam (2011) trained Artificial Neural Network (ANN) and support vector machine (SVM) using corrected SPT blow counts and cyclic stress ratio (CSR) to identify liquefaction-prone soils. In the Philippines, machine learning (ML) based models were applied by Galupino & Dungca (2023) to evaluate liquefaction susceptibility in Metro Manila from geotechnical and site-specific data. In contrast, the present study focused primarily on predicting liquefaction per 3-meter soil layer across a 15-meter profile, allowing a detailed analysis of free field soil response at different depths. This layer-based assessment provides engineers to better evaluate soil stability and identify potential hazard zones throughout the soil column.

Settlement induced by liquefaction is defined as downward displacement of the ground surface resulting from the loss of strength of soil during seismic shaking. To evaluate this behavior, Tokimatsu & Seed (1987) was identified as the most suitable empirical method for this study, as it directly uses SPT data that represents the primary geotechnical information. In contrast to Ishihara and Yoshimine (1992) and Lee (2006-2007), their method provides layer-by-layer settlement estimation for sites with multiple soil strata, ensuring that variation of free-field soil behavior with depth is captured. Additionally, the correlation of Tokimatsu & Seed (1987) is derived from large number of case histories of actual liquefaction event, thereby enhancing their reliability. Most settlement analyses are performed separately from liquefaction prediction models and rarely incorporate soil layer assessment, applying this approach across the entire soil profile provides engineers with insights into potential liquefaction-induced soil deformation, making it relevant for site evaluation.

Recent studies have demonstrated the potential of machine learning to predict liquefaction-induced settlement. For instance, Park et al. (2020) developed ANN models to estimate post-seismic settlement in Pohang, Korea using SPT-based soil parameters producing accurate layer-specific prediction. In the Philippines, machine learning applications remain limited primarily to liquefaction susceptibility mapping (Galupino & Dungca, 2023), with no study extending ML approaches to settlement estimation, particularly in a coordinate-specific manner. In contrast, the present study integrates ANN-based liquefaction prediction with depth-resolved settlement and bearing capacity analysis across a 15-meter soil profile for engineering analysis in Tarlac Province.

Bearing capacity refers to the soil's ability to support the loads from structure without experiencing shear failure. Terzhagi's (1943) classical bearing capacity theory remains the most widely adopted for estimating ultimate bearing capacity under normal, stable conditions, and used for the pre liquefaction condition. It is directly compatible with SPT-based, which forms the primary dataset of this study, and provides a reliable baseline bearing capacity before any seismic-induced strength degradation. However, Terzaghi's theory does not account for the loss of strength after liquefaction. To address this limitation, the study incorporates the post-liquefaction strength correlations developed by Olsen & Stark (2002) which offers empirical relationships for estimation of residual shear strength of liquefied soil. This approach is widely used for post-liquefaction stability assessment and is applicable for free-field soil conditions. Applying Terzaghi's theory (1943) for pre-liquefaction and Olsen & Stark (2002) for post-liquefaction conditions allows a comparative analysis of soil capacity before and after liquefaction, an approach not previously used in Philippine studies.

Local studies have provided valuable references on soil bearing capacity for engineering practice. For instance, allowable capacities across Metro Manila were estimated by Dungca et al. (2017) using corrected SPT N-values. Similarly, Lingad & Uy

(2023) applied ANN models to produce GIS maps of allowable bearing capacity in San Fernando and Santo Tomas, Pampanga. However, these studies assumed stable pre-seismic soil conditions and the effect of liquefaction on bearing capacity is not evaluated. This leaves a gap in providing engineers with information about the location-specific soil performance under earthquake-induced changes.

A foundation is a structural system that supports and distributes building loads to the ground and proper selection of foundation is critical in liquefiable soils to prevent excessive settlement and structural failure. Seed et al (2001) and Ishii and Tokimatsu (1988) establish geometric and safety thresholds required to analyze liquifiable soils, while Bray and Dashti (2010) highlighted the complex displacement mechanism that can occur during seismic events. In a study conducted by Ayadat (2017), a liquifiable site in Ontario determined that the suitable solution were either manual excavation and replacement of liquifiable soil with compacted fill or the adoption of deep piles to satisfy a 25mm settlement limit. However, current methodologies depend on manual assessment that are time-consuming and dependent on human judgement. In contrast, the present study addresses this gap by using ANN to recommend suitable foundation based on soil liquefaction potential, settlement and soil bearing capacity, providing faster and automated evaluation.

To address these gaps, this study aims to predict soil liquefaction potential, settlement estimation, pre- and post-liquefaction bearing capacity, which serve as the basis for generating appropriate foundation recommendations using ANN-based method within a unified web-based platform. Integrating these outputs into a GIS-enabled web application allows the user to input geographic locations and generate site-specific liquefaction probability maps with settlement, bearing capacity and recommends suitable foundation. This contribution strengthens academic knowledge supporting engineers and local government units, while also providing safer structural planning in liquefaction prone areas of Tarlac Province.

1.7 Hypothesis

The hypotheses were tested at a significance level of five percent (5%), which corresponds to numerical value of 0.05:

Ho: There is no significant difference in prediction accuracy between machine learning models and traditional empirical methods for assessing soil liquefaction potential, ground settlement, and soil bearing capacity.

Ha: Machine learning models show a significant difference in prediction accuracy compared to traditional empirical methods for assessing soil liquefaction potential, ground settlement, and soil bearing capacity.

1.8 Assumptions

This study aims to develop and evaluate the ANN-based liquefaction prediction system and its GIS-enabled web application. The following assumptions underpin this research:

- The ANN architecture selected is appropriate for modeling nonlinear relationships among soil parameters, liquefaction potential, settlement, and bearing capacity. This includes the model's ability to process inputs like SPT and CSR to yield reliable structural recommendation.

- The GIS-generated maps are assumed to reflect spatial tends at screening-level accuracy for preliminary planning but are not used as a total substitute for project investigation.
- Measurement and labeling errors are assumed to be random rather than systematically biased; any identified systematic errors are corrected before modeling.

1.9 Definition of Terms

The following terms, which have been included in the context of this study, are hereby defined to educate any individual with the following unfamiliar terms and to provide a deeper understanding based on its theoretical definition. The following terminologies are as follows:

1. Artificial Neural Network

   A machine learning model that can handle nonlinear and complex relationships used in classification and regression tasks.

2. Bearing Capacity

   The ability of the soil to resist vertical loads applied by a structure, specifically the foundation, before and after soil liquefaction.

3. Borehole Log

   A test result from drilling at a site that shows the vertical profile of soil types, depths, groundwater, and other geotechnical test data.

4. Soil Liquefaction

A situation in which saturated soil loses its strength during an earthquake, causing it to behave like a fluid.

5. Geographic Information System

   A visualization technology that displays the contour of each location, allowing for analysis, mapping, and sharing of geographically referenced data.

6. Spatial Visualization

   A graphical representation of data in geographic space, such as a Geographic Information System, that shows risk levels, foundation recommendations, and scenario analyses.

7. Settlement

   A rapid vertical movement of the soil caused by the rearrangement of soil particles after earthquake shaking.

Chapter 2

METHODS OF STUDY AND SOURCES OF DATA

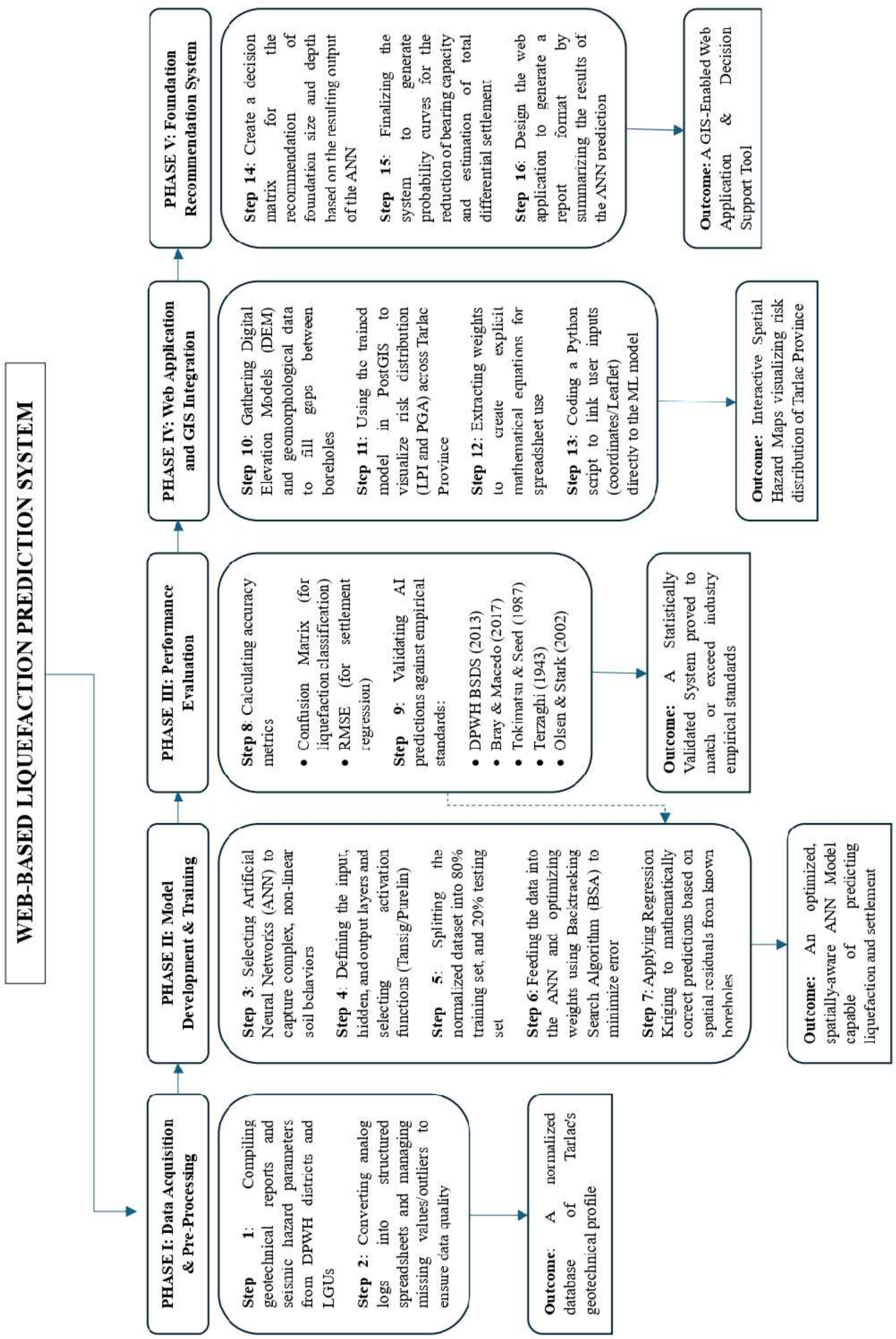
2.1 Methodology

This research uses a developmental, quantitative approach to create a web-based prediction system for assessing liquefaction potential, settlement, and bearing capacity in Tarlac Province. The study will move beyond the traditional manual calculation methods, which often struggle to account for multiple interacting variables such as fine contents, plasticity, and varying seismic intensities without oversimplification. The study combines an ANN machine learning model with geotechnical engineering principles, enabling the model to "learn" from historical borehole data patterns and leading to more accurate predictions.

The methodology outlines the phases, from research data collection and normalization to ANN model development and training using Sci-Kit Learn. The trained model was evaluated against established empirical standards – specifically the DPWH BSDS (2013) for liquefaction, Tokimatsu & Seed (1987) for settlement, Terzaghi (1943) for bearing capacity, and Olson and Stark (2002) for post-liquefaction bearing capacity. A web application was developed after evaluating the model's accuracy. Upon completion, the trained ANN model was integrated into the web application.

Figure 2

Phase Framework



Phase 1: Data Acquisition and Pre-Processing

The initial phase of the study involves compiling primary data from existing geotechnical reports, including borehole logs, SPT, seismic hazard data, and other laboratory test results. It encompasses selecting valid data from the Department of Public Works and Highways (DPWH) and Local Government Units (LGUs) in Tarlac to create a robust training dataset.

Step 1: Data Collection

Geotechnical investigation reports are compiled, specifically focusing on borehole logs, SPT blow counts (N-values), and laboratory test results such as Atterberg limits and sieve analysis. For seismic hazard assessment, data from relevant agencies—including earthquake event records, PGA values, and distance-to-fault measurements—are collected. The parameters are selected based on how significantly the data affect the target outputs, including liquefaction potential, settlement, and bearing capacity.

Step 2: Data Digitization and Cleaning

Analog borehole logs are converted into a structured spreadsheet format. Missing values and outliers are managed through synthetic data creation or correlation-based estimation of soil properties to prevent inaccuracies caused by incomplete or noisy data.

Outcome 1: A normalized database of Tarlac's geotechnical profile

Phase 2: Model Development and Training

This phase focuses on designing the machine learning framework. The goal is to create efficient algorithms that can understand complex, non-linear soil behavior, far exceeding basic regression methods. The model will be trained using the relationship between the geotechnical inputs from the previous phase and the desired outputs.

Step 3: Algorithm Selection

A suitable algorithm structure is then selected. For this study, ANN is highly recommended. It consists of several layers of interconnected computational units called neurons. These neurons allow the model to capture complex non-linear relationships (Hornik K.). To enhance the model's performance and determine the stability value (SV), the Backtracking Search Algorithm (BSA) metaheuristic is utilized.

Step 4: ANN Architecture Design

The network topology is defined, including the number of neurons in the input layer, hidden layer, and output layer to match the number of variables from the previous phase. The appropriate activation functions are selected between Tansig and Purelin to process the interactions between layers.

Step 5: Data Partitioning

The dataset is split into two sets, namely the training and test sets. This approach allows for a more comprehensive assessment of the model's validity (Choobbasti AJ, Farrokhzad F, Barari A.). In ANN forecasting models, 80% of the records are designated as training sets, while the remaining 20% are used as the testing set to monitor the model's performance during the training phase.

Step 6: Model Training

This step involved "feeding" the developed model with the dataset acquired from the previous phases to facilitate the training of the ANN. The ANN iteratively adjusts internal parameters (weights and biases) to minimize the error between predicted values and actual field data, measured using loss functions such as Mean Squared Error (MSE).

Step 7: Geostatistical Updating

The researchers incorporated regression kriging for spatial prediction. This involves using the residuals (errors) from the ML model at known borehole locations to update mathematically or correct predictions in nearby areas, ensuring the model matches observed field data where available.

Outcome 2: An optimized, spatially-aware ANN Model capable of predicting liquefaction and settlement.

Phase 3: Performance Evaluation

This phase involves validating the model's reliability through statistical and comparative analysis (geotechnical theory) to ensure it is accurate, precise, and safe for actual use.

Step 8: Statistical Analysis

After training the ML model, the model's accuracy is evaluated using metrics like Coefficient of Determination (R^2), Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Performance Index (PI). For classification models (such as determining whether the soil is liquefiable or non-liquefiable), the Confusion Matrix is employed.

$$RSME = \frac{\sqrt{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}}{\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

$$\mathrm{R}^2 = 1 - \frac{\sum_{\mathrm{i}=1}^{\mathrm{n}}(\mathrm{y_i} - \bar{\mathrm{y}}_\mathrm{i})^2}{\sum_{\mathrm{i}=1}^{\mathrm{n}}(\mathrm{y_i} - \bar{\mathrm{y}})^2}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \bar{y}|$$

Step 9: Comparative Analysis

The output is compared to the ML model's prediction with traditional empirical methods—Seed and Idriss for liquefaction, Bray & Macedo (2017) for settlement, Terzhagi or Meyerhof for bearing capacity, and Olson and Stark (2002) for post-liquefaction bearing capacity—to demonstrate superior accuracy and consistency.

Liquefaction Potential Iindex

$$LPI = \int_0^{20m} F * w(z)\, dz$$

Terzaghi's (1943) Bearing Capacity Theory

$$q_u = CN_c + \sigma_v' N_q + 0.5\gamma_{liq} BN_\gamma$$

Olsen & Stark (2002)

$$q_{u(liq)} = SrN_c + \sigma_v' N_q + 0.5\gamma_{liq} BN_\gamma$$

Outcome 3: A Statistically Validated System proved to match or exceed industry empirical standards.

Phase 4: Web Application and GIS Integration

This phase involves applying GIS in the web application. The output data from the machine learning results will be integrated with GIS software to provide a spatial visualization of the province's hazard assessment for liquefaction.

Step 10: Spatial Data Integration

The elevation models (DEM), distance to water bodies, and geomorphological maps covering the entire study area are gathered as geospatial predictor variables serving as the inputs for areas where borehole data is missing.

Step 11: Map Generation

The trained model with integrated spatial data will be used to generate hazard maps. These maps will illustrate the spatial distribution of risks, such as LPI and PGA, for specific target coordinates. The interactive map serves as the main interface of the software, displaying most of the value of the final output.

Step 12: Equation and Tool Generation

The optimized weights and biases from the ANN are extracted to develop an explicit mathematical equation. This equation allows engineers to easily determine outputs using straightforward spreadsheets without running complex code.

Step 13: Back-End Development and Integration

Using Python or Jupyter Notebook, a script is developed to build a pipeline that connects input data (Leaflet, soil data) to the ML model for quick processing. The script is designed to interact with Leaflet in .xml format, which are accessed through a user-input web address Sanger et al. (2024). Leaflet are sourced from similar global organizations, covering numerous scenario earthquakes.

Outcome 4: A functional web-based tool where users can input data and receive instant predictions, visualized on a GIS map of Tarlac.

Phase V: Foundation Recommendation System

The last phase of the study is the conversion of the complex AI model into a practical tool for engineers and producing the final study output. This is where the final interface of the web application is created.

Step 14: Logic Rule Development

Using a dual pathway analysis, a decision matrix will be formed to link the ANN outputs for Foundation Recommendation. This step involves coding the logic that compares the prediction of ANN against liquefaction potential, settlement limits, and bearing capacity requirements to automatically categorize a site into one of several foundation classes and sizes based on the risk level.

Step 15: Impact Quantification

The final output includes deliverables such as probability curves for ground failure and settlement estimates, which can be used for disaster risk reduction and land-use planning. In this paper, results are presented using either binary or numerical values of the probability of liquefaction-induced ground settlement.

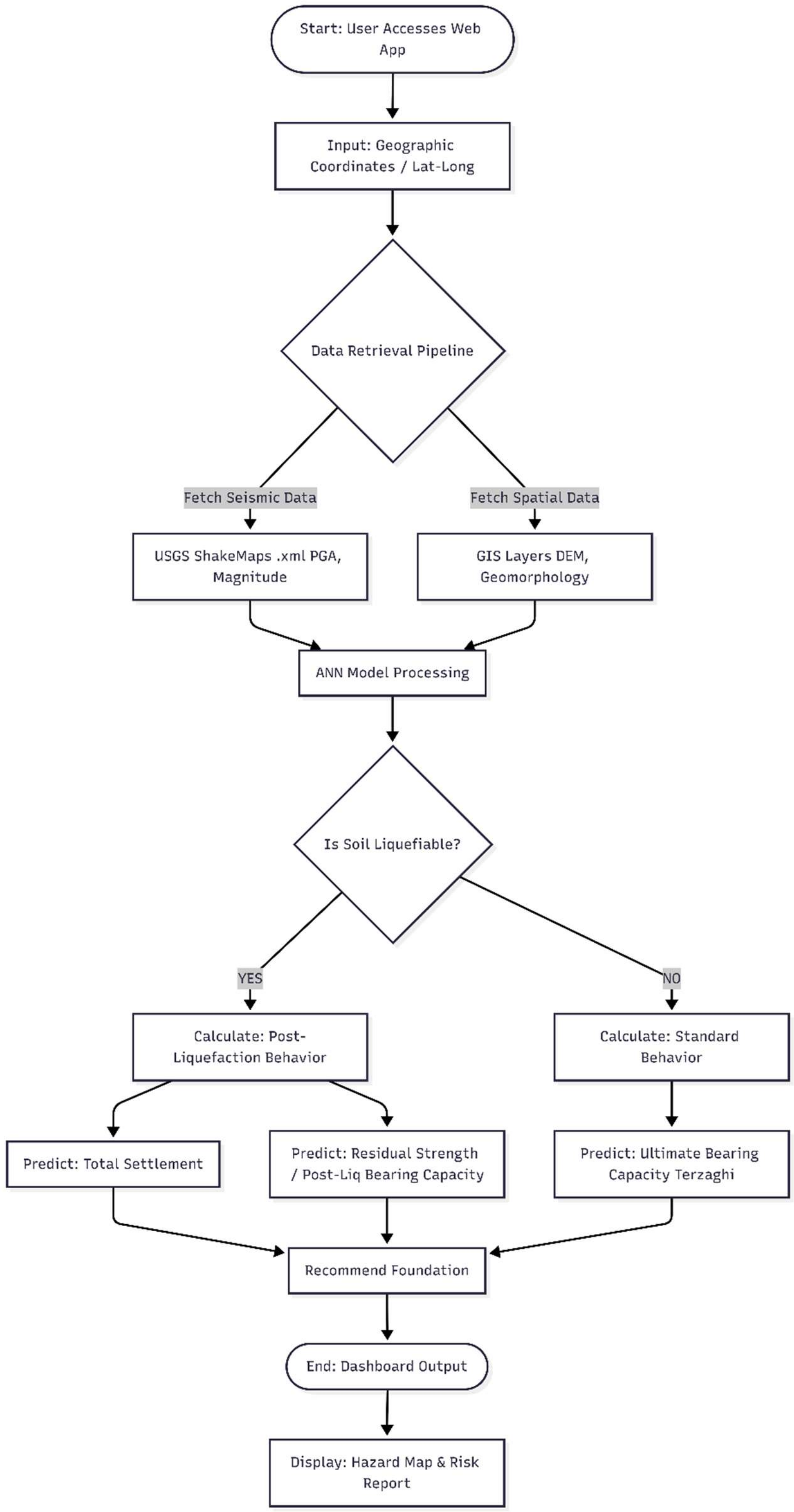
Step 16: Report Generation and Visualization

The final step is the representation of the resulting output into the web application interface. The interface will display summary report consisting of site liquefaction hazard classification, predicted values of settlement and bearing capacity, and the recommended foundation based on existing structural analysis.

Outcome 5: A GIS-Enabled Web Application & Decision Support Tool featuring Interactive Spatial Hazard Maps visualizing risk distribution.

Figure 3

Flowchart of Web-Based Liquefaction Prediction System (Tarlac Province)



2.2 Research Design

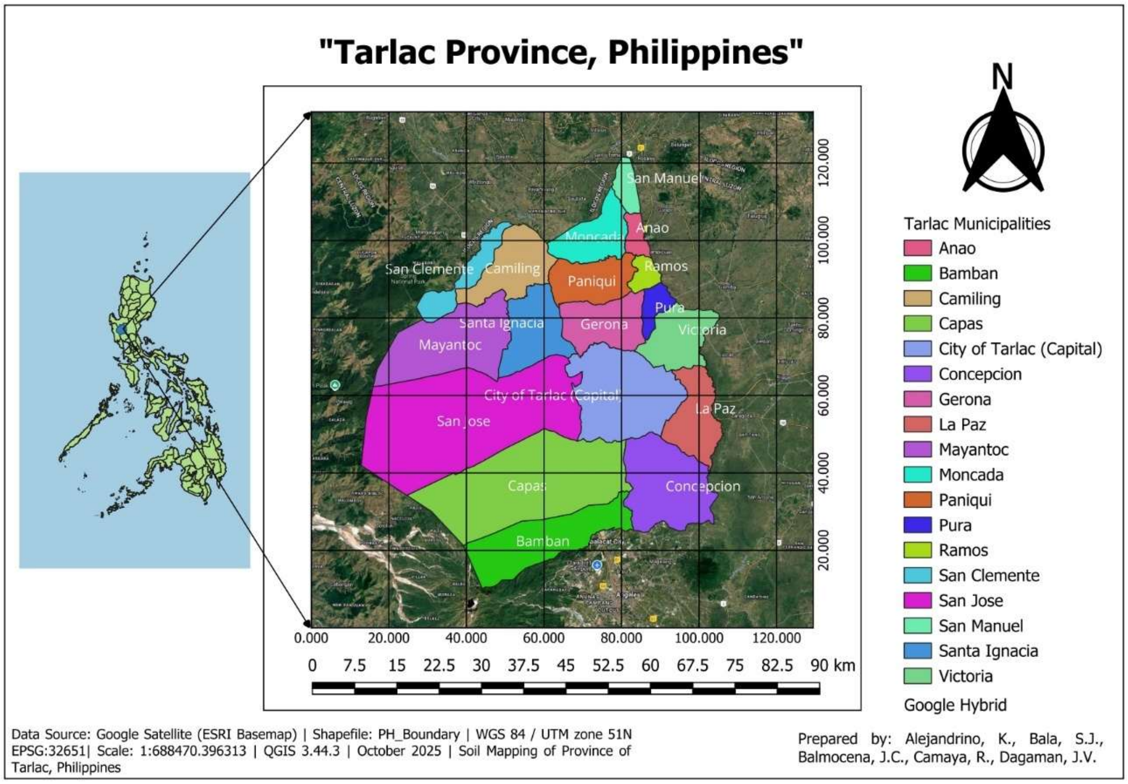
This study employs the quantitative research design to obtain its objective. Taherdoost (2021) defined a quantitative approach as one in which variables are quantified numerically to evaluate hypotheses, validate models, and produce findings that can be generalized. The primary aim of this study is to develop ANN web-based application with GIS mapping that allows users to input geographic coordinates and automatically generate soil liquefaction potential with settlement, bearing capacity and corresponding foundation recommendations. The researchers split the gathered data into two sets: the training data and the test data. The data is cut into an 80% for training and 20% for testing. The 20% testing data are validated using the Liquefaction Screening Potential in DPWH BSDS (2013), the methods of Tokimatsu & Seed (1987) and Bray & Macedo (2017) for settlement, and the methods of Terzaghi (1943) and Olsen & Stark (2002) for bearing capacity. Its performance is evaluated through Confusion Matrix Analysis for liquefaction potential and Root Mean Square Error (RMSE) for settlement and bearing capacity.

2.3 Research Setting

The study is conducted in Tarlac province, a land-locked province in the Central Luzon region which covers an area approximately 3,054 km^2. This province consists of 17 municipalities and 1 city in total, is selected for its diverse soil profiles and exposure to seismic hazards.

Figure 4

Map of Tarlac Province



2.4 Research Sample

The research sample compromises SPT-based geotechnical borehole logs from site across Tarlac Province that provide complete data required for liquefaction, settlement and bearing capacity. Researchers include only boreholes with adequate layer-by-layer data up to 15-meter depth and detailed soil profiling were included. Purposive sampling is utilized to ensure that the selected boreholes satisfied all necessary parameters for integration into the ANN for Multi-Output Regression, which predicts soil liquefaction potential, settlement, bearing capacity and produce foundation recommendations.

2.5 Data Gathering Procedures

The study implemented a comprehensive data gathering and processing framework to support the development of web-based ANN application integrated with Geographic Information System (GIS) mapping. The framework was designed to predict soil liquefaction potential, post-liquefaction settlement, and soil bearing capacity, and to generate foundation recommendation for improved structural safety in liquefaction=susceptible areas of Tarlac Province.

Geotechnical Data Collection

Geotechnical data were obtained from borehole logs and geotechnical investigation reports provided by the Department of Public Works and Highways (DPWH) Districts 1 and 2, Local Government Units (LGUs), and all municipal engineering offices within Tarlac Province. Formal requests were submitted through official letters and e-mails to ensure authorized access and data reliability.

Upon acquisition, all borehole records were reviewed to verify their accuracy, consistency, and completeness. The verification process focused on extrcating parameters essential for ANN-based prediction of liquefaction potential, settlement, and bearing capacity. The collected parameters included:

- Longitude and Latitude of boreholes
- Groundwater Level (GWL)
- Depth of borehole
- Lithology / soil stratigraphy
- Unit weight and effective unit weight

- Total and effective stress
- Standard Penetration Test corrected blow count (N_{60})
- Atterberg limits (Liquid Limit, Plastic Limit)
- Natural water content
- Mean particle size (D50)
- Fines content

Seismic Data Collection

Seismic hazard parameters were obtained from geotechnical reports and established seismic hazard references embedded in the collected datasets. These parameters included earthquake magnitude, peak ground acceleration (PGA), stress reduction coefficient, cyclic stress ratio (CSR), and cyclic resistance ratio (CRR). These variables served as critical inputs for evaluating seismic-induced soil liquefaction and its subsequent effects.

Data Processing and Organization

All verified geotechnical and seismic data were organized and encoded in Microsoft Excel. Data cleaning procedures were performed to eliminate missing values, inconsistencies, and outliers. The finalized datasets were the normalized and standardized to ensure compatibility with the ANN model. The processed data were divided into training and validation datasets to enable reliable model learning and performances assessment.

ANN Model Development, Training and Validation

A multi-output ANN model was developed to simultaneously predict liquefaction potential, settlements, and soil bearing capacity based on the processed input parameters.

Model training was performed using the training datasets, while model performance was assessed using the validation dataset.

To evaluate predictive accuracy, ANN outputs were compared with the established analytical and empirical approaches, including the DPWH Bridge Seismic Design Specifications (BSDS) Liquefaction Screening Criteria (2013), the Tokimatsu and Seed (1987) settlement method, the Bray and Macedo (2017) model, Olsen and Stark (2002) methodology, and Terzaghi's Bearing Capacity Theory (1943). These comparisons ensured the engineering reliability and developed model.

GIS-Based Spatial Integration

The validated ANN predictions are managed within PostGIS spatial databased to support efficient data retrieval and spatial analysis. To facilitate real-time visualization, these datasets are integrated into a web-based interface using Leaflet. This architecture generates interactive spatial distribution maps of liquefaction susceptibility, estimated settlement, and soil bearing capacity, allowing for dynamic and accessible hazad interpretations across Tarlac Province.

Foundation Recommendation Framework

The system incorporates a decision-support framework that generates automated foundation recommendations directly from ANN-GIS outputs. By quantifying predicted reductions in soil bearing capacity and increase in total settlement at specific coordinates, the application provides tailored engineering guidance. This framework supports safer foundation selection and informed land-use planning by translating complex geotechnical data into actionable structural recommendations.

2.6 Data Gathering Instruments

This study utilized a range of instruments and tools to support data acquisition, analysis, modeling, spatial visualization, and validation for the ANN–GIS web application.

Geotechnical Data Sources

Borehole logs and geotechnical investigation reports served as the primary instruments for subsurface characterization. These reports contained soil stratigraphy, unit and effective unit weights, SPT N_{60} values, Atterberg limits, natural water content, fines content, and particle size distribution. The data were generated through standard field and laboratory geotechnical testing procedures, ensuring dependable input data.

Seismic Data Sources

Seismic hazard parameters were derived from validated geotechnical report components and recognized seismic hazard references. These sources provided consistent values for earthquake magnitude, PGA, stress reduction coefficients, CSR, and CRR.

Mapping Tools

Spatial data management and geographic analysis are performed using a PostGIS spatial database, which serves as the backend engine for georeferenced geotechnical datasets. For the user interface, the Leaflet.js library is utilized to provide interactive mapping and real-time visualization.

Computational and Modeling Tools

Microsoft Excel was used for data encoding and preprocessing. The ANN model for multi-output regression was developed, trained, and validated using computational modeling tools. Model outputs were integrated into a web-based GIS application, enabling interactive visualization and user-friendly access to liquefaction-related predictions.

Validation Tools

The predictive performance of the ANN model was assessed using established engineering criteria and empirical methods, including the DPWH BSDS (2013) liquefaction screening guidelines, Tokimatsu and Seed (1987), Bray and Macedo (2017), Olsen and Stark (2002) methods, and Terzaghi's Bearing Capacity Theory (1943). These tools ensured the applicability of the model outputs for engineering design and hazard mitigation.

Statistical Treatment

The collected dataset is fed into an Artificial Neural Network (ANN) model. For liquefaction classification, a confusion matrix is used to evaluate the model's performance. Meanwhile, the predictions for settlement and bearing capacity are assessed using the Coefficient of Determination (R^2), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Performance Index (PI) to evaluate the performance and accuracy of the trained ANN model.

Furthermore, validation is performed by comparing ANN-generated outputs with established empirical procedures. Liquefaction classification is assessed using a DPWH Bridge Seismic Design Specifications (BSDS). Settlement predictions are compared with results derived from the Tokimatsu and Seed method, while for cases involving building-induced settlement, the Bray and Macedo (2017) method is used for validation. Bearing capacity outputs are validated against the Terzaghi formula for pre-liquefaction and Olsen

& Stark for estimated liquefaction potential. These empirical procedures are directly compared with ANN results to determine the level of agreement between the model's predictions and traditional approaches, thereby demonstrating the model's reliability and practical applicability.

2.7 Ethical Consideration

The researchers uphold strict ethical standards in conducting the study. All confidential materials, such as geotechnical data and official letters, are handled with discretion. These records are used only for research purposes and secured against unauthorized access. Identifying details are removed or anonymized in all outputs to protect the privacy of data sources.

The study maintains integrity in all stages of analysis and reporting. No data will be altered, fabricated, or misrepresented to improve the model's results. All findings will be presented transparently. Proper credit is given to all referenced works. Additionally, the study provides clear explanations for results produced by the Artificial Neural Network to ensure that the users can easily understand the logic behind each foundation recommendation.

Safety protocols are followed during data collection and technical procedures to safeguard all individuals involved. These measures ensure responsible data handling and uphold the credibility of the research.